

SAURABH KUMAR

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SUMMARY

Computer Science student with expertise in full-stack web development and artificial intelligence. Proficient in building production-ready applications using MERN stack, TensorFlow, and cloud technologies. Active open-source contributor with demonstrated success in developing scalable solutions achieving 95% model accuracy and handling 10,000+ daily API requests.

EDUCATION

Dr. APJ Abdul Kalam Technical University

Bachelor of Technology in Computer Science and Engineering; CGPA: 7.39/10.0

Lucknow, India

Aug. 2023 – May 2027

Uttar Pradesh Board (UPMSP)

Secondary School Certificate (Class X); Percentage: 87%

Uttar Pradesh, India

May 2020

TECHNICAL SKILLS

Programming Languages: JavaScript, Python, C++, HTML5, CSS3, SQL

Frontend Technologies: React.js, Vite, Tailwind CSS, Responsive Design, Single Page Applications (SPA)

Backend Technologies: Node.js, Express.js, RESTful APIs, JWT Authentication, FastApi

Databases: MongoDB, MySQL, MongoDB Atlas

AI/ML Integration: LangChain, LangGraph, Google Gemini API, TensorFlow, Keras, Playwright, RAG Pipelines, Vector Databases, Tool-calling LLMs

DevOps and Ai Tools: Git, GitHub, Docker, Vercel, Render, Jupyter Notebook, npm, Antigravity, Cursor

Core Competencies: Data Structures, Algorithms, System Design

PROJECTS

StudyMate – AI-Powered Study Platform | React, Node.js, MongoDB, Gemini AI

Jan. 2025

- Engineered full-stack educational platform serving 20+ active users with AI-powered study assistance using MERN stack and Google Gemini API.
- Implemented intelligent chatbot for personalized learning with automated document summarization supporting PDF and DOCX formats.
- Developed comprehensive analytics dashboard featuring streak tracking and progress visualization using Recharts library.
- Integrated secure OTP-based email verification system using Resend API with bcrypt password encryption and session management.
- Established DDoS protection with rate limiting (3–100 requests per endpoint) and security headers using Helmet.js middleware.
- Deployed scalable architecture on Vercel (frontend) and Render (backend) with MongoDB Atlas for reliable data persistence.
- Links: [GitHub](#) | [Live Demo](#)

Brain Tumor Detection System | Python, TensorFlow, Keras, Express.js, React, OpenCV

Dec. 2025

- Engineered full-stack medical image classification system achieving 95.59% accuracy in detecting brain tumors from MRI scans using custom CNN with Transfer Learning (VGG16).
- Integrated Express.js backend API with Python deep learning models via child processes to handle image uploads and return instant predictions.
- Designed responsive hospital portal dashboard in React featuring secure license-based registration, file uploads via Multer, and confidence score visualization.
- Preprocessed and normalized a dataset of 3,000+ brain MRI scan images using OpenCV and NumPy for model training, validation, and testing.
- Deployed production microservices architecture utilizing Vercel for the frontend and Render for the API backend.
- Links: [GitHub](#)

HavenTo – Accommodation Booking Platform | React, Node.js, MongoDB, Express

Oct. 2025

- Built full-stack booking platform handling 500+ registered users with property listings and comprehensive reservation management.

- Developed RESTful APIs processing 10,000+ daily requests with robust error handling, input validation, and response optimization.
- Implemented user authentication, favorites system, and booking functionality with secure JWT-based session management.
- Integrated Cloudinary API for optimized image storage, upload, and delivery with automatic compression and caching.
- Designed responsive user interface with React Router enabling seamless navigation and enhanced user experience across devices.
- Deployed production application on Vercel (frontend) and Render (backend) with MongoDB Atlas for scalable data storage.
- Links: [GitHub](#) | [Live Demo](#)

ACHIEVEMENTS

Contributed to 5+ open-source projects including React.js ecosystem and TensorFlow.js with merged pull requests.
Developed and deployed 3 production-ready full-stack applications on cloud platforms (Vercel, Render, AWS).
Achieved 95%+ accuracy in medical AI model for brain tumor detection from MRI scans using deep learning techniques.

Built scalable RESTful APIs handling 10,000+ daily requests with optimized performance and comprehensive error handling.
Solved 100+ algorithmic problems on LeetCode and GeeksforGeeks demonstrating strong problem-solving abilities.
Participated in college hackathons and technical competitions, securing recognition for innovative technology solutions.

CERTIFICATIONS

Front End Web Development <i>Online Course</i>	Nov. 2025
Large Language Model <i>Online Course</i>	Feb. 2026